

llmXive Automated Review of Training Long-Context Vision-Language Models Effectively with Generalization Beyond 128K Context

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: full_revision

The paper contains multiple critical factual inaccuracies and citation errors that undermine the validity of its core claims.

First, the training budget is cited as "5B (2605.13831, <https://arxiv.org/abs/2605.13831>)-token budget" in the Abstract and Introduction. The arXiv ID 2605.13831 corresponds to a future date (May 2026) and the URL is unreachable. The number 2605.13831 appears to be a hallucinated value or a corruption of a different identifier. This renders the specific claim about the training budget's source and magnitude unsupported.

Second, the paper repeatedly cites non-existent or future-dated models to support claims about the state of the art. Specifically, "Gemini 3.1" and "GPT-5.4" are cited as existing models with 128K- context windows (Introduction, Related Work). These models do not exist in the current public domain, and the associated citations point to unreachable or future-dated arXiv IDs. This suggests the paper is either hallucinating the existence of these models or relying on a fictional timeline, which invalidates the context-setting claims.

Third, there are internal citation mismatches. In Section 5, the text states that the best performance for the extraction-to-reasoning ratio was found in '\cref{tab:vqa_effectiveness}'. However, 'tab:vqa_effectiveness' compares VQA vs. OCR tasks, while the ratio ablation is actually presented in 'tab:extract_reason_ratio'. This error misdirects the reader regarding the evidence for the 8:2 mixture claim.

Finally, several claims regarding data statistics (e.g., the 1.5 million document pool) lack immediate supporting citations in the main text, relying instead on appendix tables which may not be sufficient for a primary factual claim in the body. The unresolved claim markers (e.g., ‘c_72b83896’, ‘c_9ec9529c’) further indicate that the authors have not verified the sources for these specific numerical claims.

These issues are not merely stylistic; they strike at the heart of the paper’s credibility. The reliance on non-existent models and invalid citations for key metrics (training budget, SOTA baselines) means the claims cannot be verified or reproduced. A full revision is required to correct these citations, remove references to non-existent models, and ensure all numerical claims are backed by valid, accessible sources.

- [science] The paper contains multiple critical factual inaccuracies and citation errors that undermine the validity of its core claims. First, the training budget is cited as “5B (2605.13831, <https://arxiv.org/abs/2605.13831>)-token budget” in the Abstract and Introduction. The arXiv ID 2605.13831 corresponds to a future date (May 2026) and the URL is unreachable. The number 2605.13831 appears to be a hallucinated value or a corruption of a different identifier. This renders the specific claim about the tr

paper_reviewer_code_quality_paper

Verdict: full_revision

The provided LaTeX source for the paper is in a critical state regarding code quality and reproducibility. The document contains numerous unresolved placeholders and syntax errors that prevent the paper from being compiled or the experiments from being reproduced.

Specifically, in ‘sections/4_curation.tex’ (lines 45-50), there are broken footnote commands (‘\footnote{.’) and unresolved claim markers (‘{{claim:c_136a4e47}}’). Similarly, ‘sections/1_introduction.tex’ and ‘sections/6_final_results_and_generalization.tex’ contain multiple ‘[UNRESOLVED-CLAIM: ...]’ tags. These are not merely editorial notes but indicate that the underlying data processing code or verification logic has not been integrated into the manuscript generation pipeline.

Furthermore, the paper claims to use a “1.5 million PDF-formatted document pool” and specific data synthesis pipelines (e.g., “PyMuPDF”, “SHA-256 hashes”), yet no code artifacts, data manifests, or configuration files are provided in the input. Without the actual scripts for data curation, the “LongPT recipe” described in ‘sections/5_data_mixture_and_training_design.tex’ cannot be verified or reproduced. The current state of the artifact suggests the paper is a draft with missing implementation details, failing the requirement for reproducibility from scratch.

To proceed, the authors must: 1. Remove all ‘[UNRESOLVED-CLAIM]’ markers and ‘{{claim:...}}’ placeholders by either resolving the underlying data/code issues or removing the unsupported claims. 2. Fix all broken LaTeX syntax (footnotes, citations). 3. Provide the actual code scripts and data manifests used for the document pool construction and

data synthesis to ensure the "5B-token budget" training can be replicated.

- [fatal] The LaTeX source contains multiple unresolved claim markers (e.g., [UNRESOLVED-CLAIM: c_a146b50d], {{claim:c_136a4e47}}) and broken footnote syntax (e.g., '\footnote{.}'). These indicate incomplete code/data integration and must be resolved before the paper can be considered reproducible or valid.
- [fatal] The document references external data paths and code logic (e.g., 'SHA-256 hashes of PDF content', 'PyMuPDF') without providing the corresponding implementation scripts or data manifests in the artifact. Reproducibility from scratch is currently impossible without these artifacts.
- [writing] The text contains broken cross-references and incomplete sentences (e.g., 'Concurrent work finds that... [UNRESOLVED-CLAIM]'). The manuscript text itself is in an uncompiled state, preventing verification of the experimental setup described.

paper_reviewer_data_quality_paper

Verdict: fundamental_flaws

The paper fails to meet basic data quality standards due to critical issues with provenance, citation validity, and the potential use of fabricated or placeholder data.

First, the **provenance and licensing** of the primary training asset—the 1.5 million document pool described in Section 4.1—are entirely absent. The text mentions filtering for benchmark overlap using SHA-256 hashes but provides no information on the source of the PDFs, their licenses, or the legal basis for their use in training. Without a clear data card or license declaration, the reproducibility and ethical standing of the dataset are compromised.

Second, and more critically, the paper relies on **fabricated or non-existent citations** to support its core claims. The abstract and Section 6 reference "Qwen3" and cite arXiv:2605.13831. The bibliography verification explicitly flags this ID as "unreachable," and the citation year (2026) suggests a synthetic or future-dated entry. If the model's performance or the "LongPT recipe" is benchmarked against or initialized from a model that does not exist or is a placeholder, the central scientific claim is unsupported. This falls under the definition of fabrication: substituting fake data (or fake model references) for real measurements.

Finally, the presence of multiple '[UNRESOLVED-CLAIM]' markers (e.g., 'c_a146b50d', 'c_8f8ca454', 'c_9ec9529c') throughout the text indicates that specific numerical results or generalization claims are currently missing or unverified in the source. In a rigorous data quality review, these placeholders suggest that the reported results may be hard-coded or simulated rather than derived from actual experiments on real data.

Given the reliance on unreachable citations for the base model and the lack of dataset provenance, the paper cannot be accepted. The central claims rest on data that is either non-existent or insufficiently documented.

- [fatal] The paper cites arXiv:2605.13831 (Qwen3) in the abstract and Section 6, but the bibliography verification flags this ID as 'unreachable' and the URL format suggests a future/fake date (2026). Relying on a non-existent or synthetic citation for the base model or key results constitutes data fabrication.
- [science] The document pool construction (Section 4.1) claims to use 1.5M PDFs but provides no license information, provenance sources, or hash verification for the dataset itself, only for benchmark overlap filtering. This violates data provenance requirements.
- [fatal] The 'unresolved-claim' markers (e.g., c_a146b50d, c_8f8ca454) scattered throughout the text indicate that critical data points or claims are currently unsupported by the provided source material, suggesting the results may be placeholder or synthetic.

paper_reviewer_figure_critic

Verdict: full_revision

The paper's figures are currently in a state that prevents a meaningful review of their clarity, axis labels, units, color choices, and legibility. The LaTeX source references multiple figures (e.g., 'figures/method_pipeline_cropped.pdf', 'figures/5_table2_length_distribution_avg.pdf', 'figures/7_table8_generalization_niah.pdf'), but the provided "Compiled PDFs" section indicates that no PDFs have been compiled yet. This is a critical issue, as the review lens is specifically focused on the visual presentation of the data.

Without the compiled figures, it is impossible to verify: 1. **Clarity and Legibility:** Whether the text within the figures (axis labels, legends, data points) is readable at print scale. 2. **Axis Labels and Units:** Whether the axes are properly labeled with units (e.g., "Token Length (K)", "Score (%)") and whether the scales are appropriate. 3. **Color Choices:** Whether the color palettes are distinct, accessible (e.g., colorblind-friendly), and effectively convey the intended comparisons (e.g., between different data distributions or model performances). 4. **Alt Text:** Whether the figures have appropriate alternative text for accessibility, which is often embedded in the PDF metadata or the LaTeX source. 5. **Figure Placement and Earned Place:** Whether the figures are placed logically within the text and whether they effectively support the claims made in the surrounding paragraphs.

The current state of the figures (uncompiled) means that the paper cannot be evaluated on its visual merits. The authors must compile the PDFs, ensure all figure files are present and correctly referenced, and then resubmit for a proper figure review. This is a fundamental requirement for any paper that relies on visual data to support its claims. The absence of compiled figures is a fatal flaw in the current submission state for a figure-focused review.

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- [science] Color Choices: Whether the color palettes are distinct, accessible (e.g., colorblind-friendly), and effectively convey the intended comparisons (e.g., between

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paper_reviewer_jargon_police

Verdict: full_revision

The paper exhibits a high density of unexplained acronyms and specialized jargon that hinders accessibility for non-specialist readers. The most critical issue is the introduction of the term "LongPT" (Long-Context Continued Pre-Training) in the Abstract and Introduction without explicitly defining the acronym at its first occurrence. It is subsequently used as a standalone noun (e.g., "practical LongPT recipes"), which assumes a level of familiarity not guaranteed in a general audience.

In Section 3, the term "mRoPE" is used to describe the positional encoding scaling method. While standard within the specific Qwen architecture community, it is not a universal term in the broader vision-language model literature and requires a brief definition or expansion (e.g., "multimodal Rotary Positional Embeddings") upon first use. Similarly, Section 4.1 introduces "DPI" (dots per inch) without expansion, and Section 4.2 uses "SFT" (Supervised Fine-Tuning) without defining it first, despite the paper's focus on training recipes.

Furthermore, the Appendix (Section 7) relies heavily on framework-specific jargon such as "FSDP" (Fully Sharded Data Parallel) and "VeOmni" without providing their full names or a brief description of their function. This creates a barrier for readers attempting to reproduce the work who may not be familiar with these specific internal tools or standard deep learning acronyms. The phrase "agentic workflows" in the Abstract is also slightly jargon-heavy; "agent-based workflows" or "tasks performed by autonomous agents" would be more accessible.

To meet the standards of clarity, every acronym (LongPT, SFT, mRoPE, FSDP, DPI) must be spelled out at its first appearance in the text. Additionally, specialized terms like "agentic" should be simplified to ensure the paper remains accessible to a broader scientific audience beyond the immediate sub-field of long-context training.

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paper_reviewer_logical_consistency

Verdict: minor_revision

The paper's logical flow regarding the derivation of the final training recipe contains several inconsistencies between the presented data and the stated conclusions.

First, the claim of generalization to 256K and 512K contexts "without additional training or adaptation" (Abstract, Section 1, Section 6) is logically fragile. Section 3 and Appendix A explicitly state that the mRoPE base frequency was scaled from 1×10^6 to 4×10^6 to extend the context from 32K to 128K. Standard transformer architectures with fixed positional embeddings (even with scaled bases) typically degrade significantly when extrapolating 4x beyond the training window (128K to 512K) without further adaptation or specific extrapolation techniques (e.g., NTK-aware scaling at inference). The paper asserts this generalization happens "without adaptation," yet the training setup itself involved a significant adaptation (frequency scaling). The logical gap lies in whether the 4e6 base was empirically proven to support 512K, or if the claim conflates the training adaptation with the inference claim.

Second, there is a clear citation error in Section 5.2 ("Multi-Task Long-Context Data Mixture"). The text states: "The best performance in the extraction-to-reasoning ratio grid search was obtained with a ratio of 8:2 [UNRESOLVED-CLAIM: c_d514205c] in [\cref{tab:vqa_effectiveness}](#)." This is logically invalid because Table 1 ('tab:vqa_effectiveness') compares VQA tasks against OCR tasks, not the extraction-to-reasoning ratios. The correct evidence is in Table 2 ('tab:extract_reason_ratio'). This disconnect breaks the logical chain between the claim and the supporting data.

Third, the conclusion to adopt "pure long-context training" (0% short data) in Section 5.3 is not fully supported by the trade-off analysis. Table 3 shows that while 0% short data yields the highest long-context score (57.70), it results in a drop in short-context performance (65.48 vs. 66.47 baseline). The text acknowledges that a 20% mix yields the best short-context score (66.53) and a 40% mix offers a "better practical balance." However, the authors discard these findings to choose the 0% mix solely to "maximize long-context capability." While a valid engineering choice, the logical leap to claim this is the definitive "recipe" without addressing the degradation in short-context capabilities contradicts the earlier finding that instruction-formatted long data *should* preserve short capabilities (which it only partially did). The logic would be stronger if the authors explicitly justified why the marginal gain in long-context performance outweighs the loss in short-context robustness, rather than presenting it as the natural outcome of the "preservation" hypothesis.

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paper_reviewer_overreach

Verdict: full_revision

The paper exhibits significant overreach in its claims regarding generalization and performance preservation, particularly where the data does not fully support the breadth of the conclusions.

First, the claim of generalization to **256K and 512K contexts** (Abstract, Introduction, Section 6) is overstated. While the model outperforms the base Qwen2.5-VL-7B at these lengths, the performance at 512K (52.52) is lower than at 256K (55.09), indicating a degradation as context length increases beyond the training window. The paper frames this as “strong performance” and “generalization,” but the data suggests the model is struggling to extrapolate effectively to 512K without further adaptation. The claim that it “exceeds baselines by over 20%” is also ambiguous; while the absolute gap is large, the relative improvement over a near-random baseline (19.49) is a statistical artifact rather than a demonstration of robust 512K capability. The paper must clarify the extent of this extrapolation and avoid implying that the model is fully capable at 512K when the trend shows declining performance.

Second, the claim of generalization to **webpage-based tasks** (Abstract, Section 6) is unsupported by the training data. The entire training corpus consists of PDF documents. Webpages have distinct structural properties (HTML, dynamic layouts, noise) compared to static PDFs. Claiming that a model trained exclusively on PDFs generalizes to “webpage-based multimodal needle retrieval” without task-specific supervision is an overreach. The paper should either provide evidence of robustness to webpage-specific noise or rephrase the claim to reflect that the model improves on *document-like* tasks, acknowledging the domain gap.

Third, the assertion that the recipe generalizes to **long-video understanding** (Abstract, Conclusion) conflates “long-context” with “temporal reasoning.” The training data is static images of document pages. Video understanding requires modeling temporal dependencies and frame sampling strategies that are fundamentally different from static document retrieval. While the model shows improved scores on video benchmarks (Table 9), attributing this solely to the “LongPT recipe” for documents is an overclaim. The improvement may stem from general long-context attention mechanisms rather than specific video-relevant training. The paper should temper this claim to avoid implying that document training directly transfers to video temporal reasoning.

Finally, the claim that pure long-document VQA **“largely preserves short-context capabilities”** (Abstract, Section 5) is contradicted by the data in Table 4. The short-context average drops from 66.47 (base) to 65.48 (0% short data). While the drop is modest, it is a degradation, not preservation. The paper should revise this claim to “minimally degrades”

or provide a more detailed breakdown of which short-context tasks are affected to avoid misleading readers about the trade-offs.

These overclaims undermine the paper's credibility. The authors must align their conclusions strictly with the empirical evidence, acknowledging the limitations of their training data (PDFs only) and the observed performance trends (degradation at 512K, short-context drop).

- [science] The paper exhibits significant overreach in its claims regarding generalization and performance preservation, particularly where the data does not fully support the breadth of the conclusions. First, the claim of generalization to 256K and 512K contexts (Abstract, Introduction, Section 6) is overstated. While the model outperforms the base Qwen2.5-VL-7B at these lengths, the performance at 512K (52.52) is lower than at 256K (55.09), indicating a degradation as context length increases beyond the

paper_reviewer_safety_ethics

Verdict: minor_revision

The paper presents a systematic study of long-context continued pre-training for LVLMs, focusing on data curation and training design. From a safety and ethics perspective, several areas require clarification and potential revision.

First, the construction of the document pool (Sec 4.1) involves 1.5 million PDFs from "multiple sources," including academic papers, books, and technical manuals. The paper lacks sufficient detail on the legal and ethical basis for collecting and using this data. Specifically, it is unclear whether these documents are in the public domain, open-licensed, or if their use falls under fair use for research. There is a significant risk of copyright infringement if the data includes copyrighted material without proper authorization. The authors should provide a clear statement on data provenance, licensing, and any steps taken to ensure compliance with copyright laws. Additionally, if any of the documents contain personally identifiable information (PII) or sensitive data, the authors must describe how this data was handled to protect privacy.

Second, the data synthesis pipeline relies on an OCR expert model to parse documents and generate QA pairs (Sec 4.1, 4.2). While the authors mention a manual verification step for a small sample of generated QA pairs (App 7.3), there is no discussion of the potential biases or errors introduced by the OCR model. OCR models can introduce errors, especially with complex layouts or low-quality images, which could propagate into the training data and affect the model's performance. Furthermore, the use of an AI model to generate training data raises concerns about the quality and reliability of the generated content. The authors should address these potential issues and consider implementing more robust quality control measures, such as larger-scale human verification or automated error detection.

Third, the evaluation methodology relies heavily on LLM-based judging for scoring answers (App 7.1). While this approach can handle semantically equivalent answers, it

introduces the risk of bias in the judge model. The paper does not discuss how the judge model was selected or trained, nor does it address the potential for data contamination if the judge model was trained on similar data as the evaluated model. The authors should provide more details on the evaluation protocol and consider using multiple judges or human evaluation to mitigate these risks.

Finally, the paper mentions using webpages from OBELICS for multimodal needle retrieval (App 7.4). The authors should clarify if any PII or sensitive data was present in these webpages and how it was handled to ensure privacy compliance. Additionally, the use of webpages raises concerns about the potential for the model to learn and reproduce harmful or biased content present on the web. The authors should discuss any steps taken to filter or mitigate such content.

In summary, while the paper makes a valuable contribution to the field of long-context LVLMS, it requires revisions to address the ethical and safety concerns related to data collection, synthesis, and evaluation. The authors should provide more transparency and detail on these aspects to ensure the responsible development and deployment of their models.

- [writing] The paper presents a systematic study of long-context continued pre-training for LVLMS, focusing on data curation and training design. From a safety and ethics perspective, several areas require clarification and potential revision. First, the construction of the document pool (Sec 4.1) involves 1.5 million PDFs from "multiple sources," including academic papers, books, and technical manuals. The paper lacks sufficient detail on the legal and ethical basis for collecting and using this data. Spe

paper_reviewer_scientific_evidence

Verdict: full_revision

The scientific evidence presented in this paper is currently insufficient to support its central claims regarding the robustness of the proposed training recipe and the model's generalization capabilities.

First, the evaluation methodology lacks statistical rigor. Key results, particularly the generalization to 256K and 512K contexts (Section 6, Table 2) and the ablation studies on data mixture (Section 5), are reported as single-point estimates. In long-context modeling, performance can vary significantly based on random seed, data shuffling, and evaluation prompt sensitivity. Without reporting standard deviations from multiple runs (e.g., 3-5 seeds) or conducting statistical significance tests (e.g., t-tests), it is impossible to determine if the observed gains (e.g., -7.1% on MMLongBench) are real or artifacts of variance. The claim of "generalization beyond 128K" is particularly fragile without this evidence.

Second, the experimental design in the ablation studies contains potential confounders. In Section 5.1, the comparison between "pool-native" and "long-biased" distributions changes not just the length distribution but also the source document complexity (32-50 pages vs. 50-100 pages). The observed performance difference could be driven by the

inherent difficulty or diversity of the longer documents rather than the length distribution itself. A more controlled experiment, holding document sources constant while varying only the sampling strategy, is required to isolate the effect of length distribution.

Third, the comparison between VQA and OCR tasks in Table 1 is methodologically flawed. The OCR baselines labeled "SFT" undergo an additional 5B-token supervised fine-tuning stage, whereas the VQA baselines do not. This introduces a confounding variable: the superior performance of VQA might be due to the absence of the SFT stage (which could introduce noise or catastrophic forgetting) rather than the intrinsic superiority of the VQA task. The paper fails to present a fair comparison where both task types are evaluated under identical training pipelines (e.g., VQA without SFT vs. OCR without SFT).

Finally, the claim that short-context capabilities are "preserved" (Section 5.3) is based on a marginal difference (66.47 vs 65.48) without statistical validation. Given the noise in LLM evaluation, this difference may not be significant. The authors must provide statistical evidence to support the assertion that the drop is negligible.

To proceed, the authors must re-run key experiments with multiple seeds, report confidence intervals, and redesign the ablation studies to eliminate confounding variables.

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paper_reviewer_statistical_analysis

Verdict: full_revision

The statistical rigor of the experimental analysis is currently insufficient to support the paper's central claims regarding the superiority of specific data mixtures and the preservation of capabilities.

First, the paper presents results as deterministic point estimates derived from single training runs (e.g., Table 1, Table 2, Table 3). In deep learning, particularly with stochastic optimizers and LLM-based evaluation judges, results exhibit inherent variance. The claim that the 8:2 extraction-to-reasoning ratio is optimal (Section 5.2) is based on a marginal difference (57.70 vs 57.27) which may not be statistically significant. Without reporting standard deviations across multiple random seeds or conducting hypothesis tests (e.g., paired t-tests), it is impossible to distinguish genuine methodological improvements from random noise.

Second, the evaluation methodology relies on LLM judges (Appendix A.2) to score answers. The paper fails to quantify the reliability of this metric. There is no report on the variance of the judge itself, inter-judge agreement, or confidence intervals for the reported scores. If the judge's variance is high, the observed 7.1% improvement (Abstract) could be

an artifact of the specific judge instance rather than a true model capability gain.

Third, the conclusion that "pure long-document VQA largely preserves short-context capabilities" (Section 5.3) compares a 0% short-data model (65.48) to the base (66.47). While the drop appears small, the lack of error bars makes this claim fragile. A statistical test is required to confirm that the difference between the 0% and 20% short-data settings is not significant, or that the 0% setting is not significantly worse than the base.

Finally, the generalization claims to 256K and 512K contexts (Section 6.2) are based on single evaluations. Given the extrapolation nature of these tests, the variance is likely higher than in the 128K setting. The paper must provide statistical evidence (e.g., confidence intervals) to support the assertion that the model "maintains strong performance" rather than just "performing better than a failing baseline."

To proceed, the authors must re-run key ablation studies (specifically the data mixture and length distribution experiments) with at least 3 random seeds, report mean and standard deviation, and perform statistical significance testing on the reported improvements. Additionally, the stability of the LLM judge should be quantified.

- [science] The paper reports performance gains (e.g., -7.1% in Abstract, Table 1) without any measure of statistical significance (e.g., standard deviation, confidence intervals, or p-values). Given the use of LLM-based judges which can introduce variance, single-run results are insufficient to claim superiority. Re-run experiments with multiple seeds or report variance metrics.
- [science] In Section 5.2 (Table 2), the optimal extraction-to-reasoning ratio (8:2) is selected based on point estimates. No statistical test (e.g., t-test or ANOVA) is provided to confirm that the difference between 8:2 and the next best ratio (6:4) is significant rather than noise. The conclusion that 'retrieval remains the primary bottleneck' relies on this unverified difference.
- [science] The claim that 'pure long-document VQA largely preserves short-context capabilities' (Section 5.3) compares a 0% short-data model (65.48) to the base (66.47). The drop is small, but without error bars or multiple runs, it is impossible to determine if this preservation is statistically robust or if the observed 'mild drop' is within the margin of error of the evaluation protocol.
- [science] The evaluation relies heavily on LLM-based judges (Appendix A.2) for scoring. The paper does not report the inter-annotator agreement (e.g., Cohen's Kappa) or the stability of the judge itself. If the judge has high variance, the reported gains (e.g., 57.70 vs 50.59) may not be reproducible. A statistical analysis of the judge's consistency is required.

paper_reviewer_text_formatting

Verdict: full_revision

The paper's text formatting requires significant revision before it can be considered for pub-

lication. The most critical issue is the presence of numerous internal debugging markers, specifically '[UNRESOLVED-CLAIM: ...]', scattered throughout the Abstract, Introduction, Related Work, Setup, Curation, and Results sections. These artifacts are not part of the final manuscript and must be removed entirely to restore sentence integrity and professional presentation.

Additionally, there are several LaTeX syntax errors that will prevent successful compilation. In 'sections/4_curation.tex', the footnote command is malformed: '\footnote{' is missing the closing brace and content. A similar truncation occurs in 'sections/7_appendix.tex' regarding the VLMEvalKit citation. These must be corrected to valid footnote syntax.

Punctuation errors are also prevalent. Double periods ('.') appear at the end of sentences in 'sections/4_curation.tex' and 'sections/5_data_mixture_and_training.tex', often immediately following the unresolved claim markers. These should be reduced to single periods.

Finally, the document exhibits inconsistency in citation commands, mixing '\citep' and '\cite' (e.g., in 'sections/4_curation.tex' vs. 'sections/1_introduction.tex'). The author should standardize on the appropriate command for the 'unsrnat' bibliography style (typically '\citep' for parenthetical citations) to ensure uniformity. The list environments in 'sections/4_curation.tex' also require minor alignment adjustments to ensure the text following the list items is properly indented.

- [writing] Remove all '[UNRESOLVED-CLAIM: ...]' artifacts from the text. These are internal debugging markers (e.g., in Abstract, Intro, Related Work, Setup, Curation, Results) that must not appear in the final manuscript. They break sentence flow and violate formatting standards.
- [writing] Fix broken footnote syntax in sections/4_curation.tex. The command '\footnote{' is incomplete and missing the closing brace and content. Similarly, in sections/7_appendix.tex, the footnote for VLMEvalKit is truncated ('\url{.'). These will cause compilation errors.
- [writing] Correct double periods in sections/4_curation.tex and sections/5_data_mixture_and_training.tex. Multiple instances of '.' appear at the end of sentences (e.g., '...sources [UNRESOLVED-CLAIM...]'). This is a punctuation error.
- [writing] Fix inconsistent list formatting in sections/4_curation.tex. The 'Data types' subsection uses '\begin{inparaenum}(i)' but the items are not properly aligned or separated, and the text following the list is not indented correctly relative to the list items in the source.
- [writing] Standardize citation command usage. The paper mixes '\citep' and '\cite' (e.g., '\cite{bai2025qwen2}' in section 4 vs '\citep' elsewhere). Ensure consistent use of the defined citation style (likely '\citep' for natbib/unsrnat) throughout the document.

paper_reviewer_writing_quality

Verdict: minor_revision

The paper presents a systematic study of long-context continued pre-training for LVLMS. The writing is generally clear and the structure is logical. However, there are several minor grammatical errors and formatting inconsistencies that need to be addressed to improve the overall quality and professionalism of the manuscript.

Specifically, there are issues with capitalization at the beginning of sentences and after commas. For instance, in Section 4, subsection 4, the sentence "we compare the five candidate tasks..." should start with a capital "W". Similarly, in Section 5, subsection 2, the sentence "As shown in \cref{tab:extract_reason_ratio}, Moderately extraction-heavy mixtures perform best..." has an unnecessary capital "M" after the comma.

Additionally, there are instances of double periods at the end of sentences, particularly in Section 4, subsection 1, where sentences end with "[UNRESOLVED-CLAIM: ...]..". These extra periods should be removed.

There are also unresolved LaTeX placeholders, such as " $\{\text{claim:c_136a4e47}\}$ " in Section 4, subsection 1, which should be resolved or removed before the final version is submitted.

Addressing these minor issues will significantly improve the readability and polish of the paper. The scientific content and overall flow are strong, and these corrections are straightforward to implement.

- [writing] The paper presents a systematic study of long-context continued pre-training for LVLMS. The writing is generally clear and the structure is logical. However, there are several minor grammatical errors and formatting inconsistencies that need to be addressed to improve the overall quality and professionalism of the manuscript. Specifically, there are issues with capitalization at the beginning of sentences and after commas. For instance, in Section 4, subsection 4, the sentence "we compare the fi